

Math723 – Differential Equations I

Fall 2014

Final Exam

You must show all of your work to get a credit for a correct answer.

1. Let $u(x, y)$ be a harmonic function in $\mathbb{R}^2$ and suppose that

$$\iint_{\mathbb{R}^2} |\nabla u|^2(x, y) \, dx dy < \infty.$$

Prove that u is a constant function.

(Hint: Use the property that $|\nabla u|$ is subharmonic.)

2. Let u be a positive harmonic function in the unit ball $B(\mathbf{0}, 1)$ in $\mathbb{R}^n$. Show that

$$|D(\ln(u))| \leq M, \quad \text{in } B(\mathbf{0}, 1/2),$$

for some constant M depending only on the dimension n .

(Hint: Use the interior derivative estimate for $|Du|$ on $B(\mathbf{0}, r) \subset B(\mathbf{0}, 1)$ and the Harnack inequality for harmonic functions.)

3. Find the solution of the initial value problem

$$\begin{cases} u_{tt} - u_{xx} + u &= 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, 0) &= f(x), & \text{in } \mathbb{R}, \\ u_t(x, 0) &= g(x), & \text{in } \mathbb{R}, \end{cases}$$

where $f, g \in C_0^\infty(\mathbb{R})$.

(Hint: Let $v(x, y, t) = h(y)u(x, t)$ and find h so that v solves the two-dimensional wave equation.)

4. Solve using characteristics

$$\begin{cases} u_t - x_2 u_{x_1} + x_1 u_{x_2} &= 0, & \text{in } \mathbb{R}^2 \times (0, \infty), \\ u(x_1, x_2, 0) &= e^{-x_2^2}, & \text{in } \mathbb{R}^2, \end{cases}$$

5. Consider the conservation law

$$u_t + f(u)_x = 0, \quad \text{in } \mathbb{R} \times (0, \infty),$$

where $f(u) = u^2 + u$. Find a piecewise smooth integral solution of the equation with

$$u(x, 0) = \begin{cases} 1, & x < 0, \\ -3, & x > 0. \end{cases}$$